



IITM

MAD – I Project

----- Quiz Management App -----

Created By:

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Executive Summary :

The Ischool Quiz App is a robust, scalable Quiz Management System, designed to optimize the creation, deployment and evaluation of quizzes. Developed utilizing concepts from Modern App Development – 1, this app ensures seamless user experience for both administrators and participants, enhancing the efficiency of online assessment.

Technology Stack:

- **Backend:** Flask (Python) with Flask-SQLAlchemy for ORM-based database interaction.
- **Database:** SQLite for structured, lightweight data persistence.
- **Frontend Layer:** Jinja2 templating engine combined with Bootstrap for a responsive and professional UI.

Key Frameworks and Libraries:

- **Flask-SQLAlchemy** – Object Relational Mapping for streamlined database management.
- **Flask Core Modules:**
 - **render_template:** Generates dynamic HTML templates.
 - **request:** Captures and processes user inputs.
 - **redirect:** Manages internal navigation workflows.
 - **url_for:** Ensures dynamic and maintainable URL routing.
 - **flash:** To sending message from backend.
- **Bootstrap & Jinja2** – Enables a modern, user-friendly interface with responsive design elements.

Core Functionalities and Features:

Administrative Capabilities:

1. Full CRUD operations for subject, chapter, quiz, and question management.
2. Advanced analytics dashboard featuring real-time user engagement insights and quiz performance tracking.
3. User Management: Ability to activate, suspend, or delete users to maintain platform integrity.
4. Enhanced search and filtering, enabling administrators to retrieve specific records efficiently.

End-User Functionalities:

1. Secure Authentication & Authorization – Users register and authenticate using a robust credential system.
2. Personalized Dashboard displaying performance metrics, past quiz attempts.
3. Real-time Quiz Participation with an integrated timer and answer validation
4. Post-quiz analytics, including score breakdowns, correctness analysis, and improvement suggestions.
5. Comprehensive Query System – Users can search for quizzes by subject, chapter, date, and performance scores

Operational Workflow:

Administrative Process Flow

1. **Dashboard Access:** Admins are directly redirected to a secure control panel.
2. **Quiz Configuration:** Creation of subjects, chapters, and quizzes with custom parameters.
3. **Question Bank Management:** Definition of multiple-choice questions with associated correct responses and timers.
4. **User & Data Governance:** Performance monitoring, user access management, and report generation.

User Engagement Process

1. **Authentication & Access:** Users log in securely via the authentication system.
2. **Quiz Selection & Participation:** Users browse available quizzes and engage in real-time assessment.
3. **Answer Submission & Validation:** Responses are recorded, evaluated, and scored.
4. **Performance Review:** Users receive comprehensive results, including correctness breakdowns and improvement suggestions.

Database Schema

All Database schema is given shows in pictorial form.

